

Original Report

A choice of radionuclide: Comparative outcomes and toxicity of ruthenium-106 and iodine-125 in the definitive treatment of uveal melanoma



Vinita Takiar MD, PhD^a, K. Ranh Voong MD^a, Dan S. Gombos MD^b,
Firas Mourtada PhD^{c,d}, Laura A. Rechner MS^d, Ann A. Lawyer MS^d,
William H. Morrison MD^a, Adam S. Garden MD^a, Beth M. Beadle MD, PhD^{a,*}

^aDepartment of Radiation Oncology, University of Texas MD Anderson Cancer Center, Houston, Texas

^bDepartment of Ophthalmology, University of Texas MD Anderson Cancer Center, Houston, Texas

^cDepartment of Radiation Oncology, Helen F. Graham Cancer Center, Christiana Care Health System, Newark, Delaware, and Thomas Jefferson University, Philadelphia, Pennsylvania

^dDepartment of Radiation Physics, University of Texas MD Anderson Cancer Center, Houston, Texas

Received 24 July 2014; revised 5 September 2014; accepted 17 September 2014

Abstract

Purpose: Both iodine-125 (¹²⁵I) Collaborative Ocular Melanoma Study and ruthenium-106 (¹⁰⁶Ru) eye plaques can achieve excellent tumor control in patients diagnosed with uveal melanoma. We analyzed our single institutional experience in the management of ocular melanoma treated with either ¹²⁵I or ¹⁰⁶Ru plaque brachytherapy.

Methods and materials: The records of 107 patients with uveal melanoma treated with either ¹⁰⁶Ru (n = 40) or ¹²⁵I (n = 67) plaque brachytherapy between 2000 and 2008 were retrospectively reviewed. Tumor control parameters and toxicity were assessed.

Results: Actuarial 5-year rates of local control, progression-free survival, and overall survival with ¹⁰⁶Ru were 97%, 94%, and 92%, respectively. For ¹²⁵I, these values were 83%, 65%, and 80%. In the subset of patients with tumor apex height ≤ 5 mm (36 ¹²⁵I and 40 ¹⁰⁶Ru), there was no difference in overall survival; however, progression-free survival was significantly improved with ¹⁰⁶Ru (*P* = .02). Enucleation-free survival was significantly different between the 2 subsets, with no enucleations in the ¹⁰⁶Ru cohort (*P* = .02). Patients treated with ¹⁰⁶Ru experienced reduced retinopathy (*P* = .03) and cataracts (*P* < .01).

Conclusions: Both ¹²⁵I and ¹⁰⁶Ru eye plaque brachytherapy treatment result in encouraging tumor control for patients with uveal melanoma. We demonstrate that ¹⁰⁶Ru offers these benefits with reduced toxicity in patients treated for uveal melanomas ≤ 5 mm in apical height.

© 2015 American Society for Radiation Oncology. Published by Elsevier Inc. All rights reserved.

The abstract of this study was presented at the 55th Annual Meeting of the American Society for Radiation Oncology, Atlanta, GA, September 22–25, 2013.
Conflicts of interest: None.

* Corresponding author. 1515 Holcombe Blvd, Unit 0097, University of Texas MD Anderson Cancer Center, Houston, TX 77030.
E-mail address: bbeadle@mdanderson.org (B.M. Beadle).

<http://dx.doi.org/10.1016/j.prro.2014.09.005>

1879-8500/© 2015 American Society for Radiation Oncology. Published by Elsevier Inc. All rights reserved.

Introduction

Uveal melanoma is the most common primary intraocular malignancy in adults, with an age-adjusted incidence of 5.1 per million.¹ Uveal melanomas comprise 3.1% of all melanomas.² The mean age at diagnosis is 58 years.²⁻⁴ Most patients become symptomatic from their tumors, presenting with visual loss, pain, or pressure. As such, uveal melanoma is usually diagnosed before disease dissemination, allowing for definitive local therapy.^{5,6} Five-year overall survival (OS) is reported to be in the range of 69%-83%.^{7,8}

Management options for uveal melanoma depend on the size and location of the tumor. Historically, enucleation was the standard of care. However, randomized trials demonstrated that organ preservation with definitive radiation therapy did not alter the risk of melanoma-related mortality. Therefore, radiation therapy has become the preferred treatment for patients with small- and medium-sized tumors (apical height 2.5-10 mm).¹ Treatment with radiation allows for the possibility of visual preservation and improved cosmesis. The original experience in the use of radiation for definitive treatment was with cobalt-60; however, more recent experiences, including the Collaborative Ocular Melanoma Study (COMS) trial, used iodine-125 (¹²⁵I) in an attempt to minimize toxicity.^{9,10}

The phase 3 COMS trial compared enucleation with ¹²⁵I brachytherapy for the treatment of medium-sized tumors. There was no difference in melanoma-related mortality noted in the 2 arms of the study; this established ¹²⁵I brachytherapy as standard of care for the treatment of medium-sized intraocular melanomas.¹¹⁻¹³ ¹²⁵I is a low-energy gamma ray emitter with 28.4 keV mean energy and a 59.4 day half-life.^{14,15} One significant concern of using ¹²⁵I plaques has been the potential toxicity given the depth of penetration of this radionuclide compared with low-energy beta-emitting radionuclides. Clinically significant deterioration in visual acuity has been observed in 42%-49%

of study patients treated with ¹²⁵I approximately 3-4 years after treatment.^{16,17} In addition, episcleral ¹²⁵I plaque brachytherapy, at times, requires disinsertion of the extraocular muscles for proper placement because of the thickness of the plaque itself.

As the treatment focus has shifted to decreasing radiation-related complications and maintaining visual acuity, other radionuclides have also been used in the definitive treatment of uveal melanoma. ¹⁰⁶Ru has an approximately 372 day half-life and decays into rhodium-106 with beta-particle maximum and mean energy of 3.54 MeV and 1.42 MeV, respectively. ¹⁰⁶Ru has a lower depth of penetration than ¹²⁵I and has been studied for the therapy of melanoma tumors ≤ 5 mm in apical thickness in the United States.¹⁸ Commercially available ¹⁰⁶Ru plaques are notably thinner than the corresponding sized COMS ¹²⁵I plaques, because ¹⁰⁶Ru is electrodeposited on the surface of a 0.2-mm-thick silver substrate mounted on a 0.7-mm silver backing and covered with a 0.1 mm thick silver window (Fig 1). These thinner plaques facilitate placement without the need to disinsert extraocular muscles. There are neither published prospective trials nor single institutional retrospective comparisons of tumor control or toxicity outcomes with these 2 radionuclides in patients with uveal melanomas ≤ 5 mm in apical height.

The goals of the current study are to establish the long-term outcomes and toxicity profiles of definitive brachytherapy treatment with ¹²⁵I for ocular melanoma in the modern era. In addition, we compare the subset of patients treated with ¹²⁵I that were also appropriate for ¹⁰⁶Ru (namely, patients with tumors with an apical height of ≤ 5 mm) with a similar cohort treated with ¹⁰⁶Ru, allowing us to test the preliminary hypothesis that outcomes with ¹⁰⁶Ru may be similar, if not superior to ¹²⁵I for this patient population. For the first time, there is a direct comparison of disease outcomes and toxicity between the radionuclides in a similar group of patients, treated at the same institution, by the same physicians, and with the same

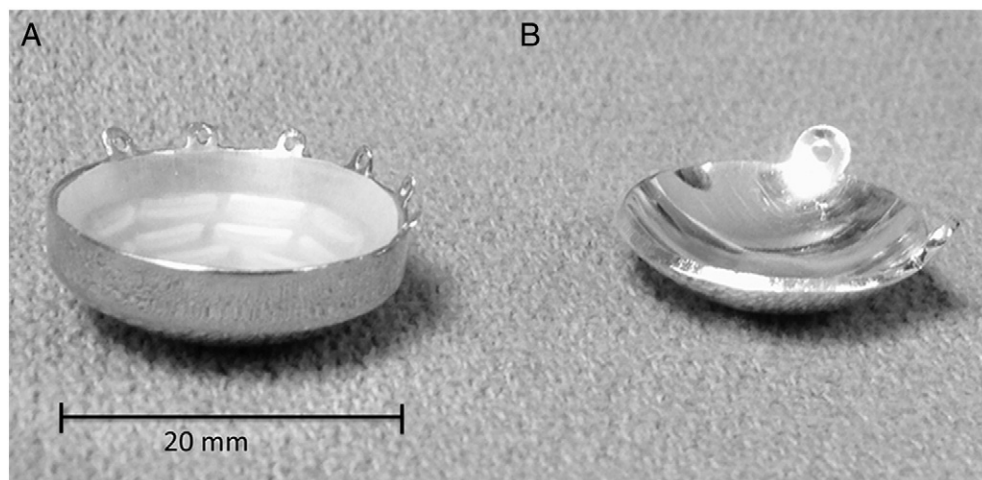


Figure 1 (A) A 20-mm iodine-125 plaque. (B) A 20-mm ruthenium-106 CCB plaque.

standard evaluations and follow-up schema. This analysis is especially timely because ^{106}Ru had been removed from the United States market for many years and has recently been reintroduced; physicians now have a choice of radionuclide in the treatment of their patients with ocular melanomas (≤ 5 mm in apical height).

Methods and materials

Patients

We conducted a retrospective review of patients treated with ^{106}Ru and ^{125}I eye plaque brachytherapy for uveal melanoma between 2000 and 2008. Our ^{106}Ru use commenced in 2003 and was discontinued in 2008 as the radionuclide was removed from the US market at that time. Therefore, ^{125}I was reinstituted for patients who were formerly appropriate candidates for ^{106}Ru plaque brachytherapy. This retrospective study was approved by our Institutional Review Board.

Detailed descriptions and analyses of patients treated with ^{106}Ru have been previously described.¹⁹ Briefly, 40 patients were treated with ^{106}Ru using commercially available eye plaques (Eckert & Ziegler BEBIG GmbH, Berlin, Germany) with plaque size and type determined by the treating physicians based on the tumor dimensions and proximity to the optic nerve. Seventy-one patients were identified who were treated with ^{125}I eye plaque brachytherapy (seed model 6711, GE Healthcare, Arlington Heights, IL) from 2000 to 2008. Four of these patients had metastatic disease at the time of initial ophthalmologic evaluation and were therefore excluded from further analysis. The remaining 67 patients represent the ^{125}I study population. Commercially available COMS plaques (Trachsel Dental Studio, Rochester, MN) were used after proper dosimetric quality assurance per American Association of Physicists in Medicine TG-129 recommendation, with the type and diameter of plaque determined by the size and the location of the lesion.²⁰

Data analysis

Patients were collectively analyzed for tumor control parameters including OS, local control (LC), progression-free survival (PFS), and enucleation-free survival (EFS). OS is defined as the percentage of patients alive at the designated time. LC implies that the patient has no evidence of intraocular recurrence at follow up. Distant metastasis-free survival is the time period until evidence of distant disease manifests. PFS implies that the patient is alive without any evidence of disease progression. Disease progression in the treated eye was defined as an increase in size on clinical ophthalmoscopy, ocular echography, or fundus photography. All of the patients in this study (receiving ^{106}Ru and ^{125}I brachytherapy) were treated and

followed by the same ophthalmologist (DG), allowing for internally consistent documentation of ophthalmologic findings. Regular screening for distant metastatic disease was performed every 6 months for the first 5 years and then annually thereafter. Additionally, if a patient was known to have class II melanoma, liver function testing was performed every 3 months.

Of the 67 patients treated with ^{125}I , plaque size information was available for 65 patients, dose rate was available for 62 patients, inner scleral dose and opposite retina dose information were available for 66 patients, and treatment time information was available for 62 patients. For subset analysis, scleral dose and opposite retinal dose information was available for 35 patients and dose rate at prescription depth for 33 of the 36 patients. Analysis was performed on the available data.

To assess for changes in visual acuity, treated patients were subdivided into predefined groups (20/20 up to 20/40, 20/40 up to 20/200, and worse than 20/200) based on preimplant and postimplant visual acuity in the affected eye. Movement between these visual acuity groups was characterized as clinically significant visual loss. Cataracts were noted if they were clinically significant. Radiation-induced toxicity was assessed by the treating ophthalmologist for 66 of the 67 patients treated with ^{125}I because 1 patient (with tumor height ≤ 5 mm) died in the immediate postoperative period after plaque removal resulting from cardiac arrhythmia. Thirty-five patients who had a tumor height ≤ 5 mm treated with ^{125}I were compared in subset analysis with all 40 of the patients treated with ^{106}Ru (all with height ≤ 5 mm). Comparison of toxicities was performed using a chi-square test with P value $< .05$ and 1 degree of freedom. All statistics were calculated using SPSS Statistics 21 (IBM Corporation).

Results

Patient characteristics

Details regarding patients treated with ^{106}Ru were published previously.¹⁹ Of the 67 patients treated with ^{125}I , 1 had iridociliary melanoma, 3 had ciliochoroidal melanoma, and the remaining 63 patients were identified as having choroidal melanoma. Each patient had only 1 affected eye and received 1 implant. Thirty-five patients presented with disease in the right eye and 32 in the left eye. Thirty-five patients were female; 32 were male. In addition to brachytherapy, some patients received additional adjuvant therapy, including: transpupillary thermotherapy (31 patients), laser photocoagulation (15 patients), Avastin (6 patients), and cryotherapy (3 patients), as deemed appropriate by the treating ophthalmologist at the time of the procedure. Median age at the time of implant was 62 years (range, 37-95).

Tumor and plaque characteristics

The median apical height of tumors treated with ^{106}Ru was 3.05 mm; the median of the maximal radial dimension was 9.6 mm. Median prescribed dose to the apex was 90 Gy.¹⁹ Table 1 details baseline tumor and plaque characteristics of patients treated with ^{125}I . Nineteen patients were treated with notched plaques, which were preferred when the lesion was within 2 mm of the optic nerve. Plaque sizes were 14 mm (2 patients), 16 mm (9 patients), 18 mm (21 patients), 20 mm (23 patients), and 22 mm (10 patients). Plaque size could not be verified for 2 patients. In accordance with dosing in the COMS trial, 32 of the 36 ^{125}I patients with lesions <5 mm in apical height had dose prescribed to 5 mm with a median prescription dose of 85 Gy.¹² The other 4 patients were prescribed a median of 85 Gy to 1 mm beyond the apical height of their tumors (3, 4, 4, and 4 mm, respectively).

Overall, the median prescribed apical dose for all 67 patients treated with ^{125}I was 85 Gy. Median dose rate at prescription depth was 13 mGy/min. Median inner scleral dose ranged from 208 Gy to 678 Gy. Opposite retina dose received ranged from 5.1 Gy-25 Gy. Treatment time ranged from 73-160 hours with a median of 121 hours.

Disease control

Median ophthalmologic follow up for patients treated with ^{125}I was 61 months (range 0-132). For ^{125}I patients, the actuarial 2- and 5- year overall survival rates were 93% and 80%, respectively (Fig 2A). Twenty-two deaths occurred, including 1 patient who died of cardiac arrhythmia in the immediate postoperative period after plaque removal. Actuarial LC was 91% at 2 years and 83% at 5 years (Fig 2B). Actuarial 2- and 5-year PFS was 85% and 65% (Fig 2C). Distant metastases occurred in 18 patients for actuarial distant metastasis-free survival rates of 88% and 71% at 2 and 5 years. There were a total of 9 enucleations for a 5-year EFS of 87% (Fig 2D). Four

enucleations were due to local progression, 2 were due to progressive neovascular glaucoma and resultant pain, 1 was due to vitreous hemorrhage, 1 was due to suspicion of hemorrhage at an outside hospital, and 1 was performed several weeks postimplant because of retinal detachment with no light perception in the treated eye. The median apical tumor height of the 9 patients who were treated with enucleation was 4.55 mm (range 2.8-10.8 mm). Disease control and ocular preservation outcomes for patients treated with ^{125}I are detailed in Table 2.

Comparison of outcomes in patients with tumors ≤5 mm in height

The subset of patients treated with ^{125}I with apical tumor height ≤5 mm was compared with patients treated with ^{106}Ru , all of whom had an apical tumor height ≤5 mm. There were 9 deaths and 10 patients with disease progression in the subset treated with ^{125}I , and 3 deaths and 2 patients with progression in the ^{106}Ru cohort. By log-rank analysis, there was no difference in OS ($P = .21$), however PFS was significantly increased ($P = .02$) in the subset of patients treated with ^{106}Ru as depicted in Fig 3A and C, respectively. There was also a trend toward improved LC with ^{106}Ru ($P = .07$) as depicted in Fig 3B, with only 1 local failure in the ^{106}Ru population in comparison to 5 local failures in the ^{125}I -treated subset. There was a statistically significant difference in EFS favoring the group treated with ^{106}Ru , with 5 enucleations in the ^{125}I -treated subset and zero in the ^{106}Ru group ($P = .02$) (Fig 3D).

Radiation-associated toxicity

Forty-three patients treated with ^{125}I (65%) experienced radiation retinopathy; 31 (47%) were diagnosed with a cataract related to their eye plaque brachytherapy, 4 (6%) were diagnosed with neovascular glaucoma attributed to their treatment, 5 (8%) experienced posterior vitreous

Table 1 Tumor characteristics and plaque dosimetric parameters

Parameter	Median value (range)		
	I-125 patients (N = 67)	Ru-106 patients (N = 40)	I-125 patients with tumor height ≤5 mm (N = 36)
Apical height (mm)	4.94 (1.94-12.06)	3.05 (1.27-4.94)	3.65 (1.94-4.98)
Maximum radial diameter (mm)	12 (4.85-18)	9.6 (3.1-13.0)	10.3 (4.85-18)
Apical dose (Gy)	85 (62-140)	90 (85-90)	104 (80-140)
Prescription dose (Gy)	85 (75-140)	90 (85-90)	99 (80-140)
Inner sclera dose (Gy)	279 (208-678)	252 (156-453)	260 (208-481)
Opposite retina dose (Gy)	8.7 (5.1-25.0)	Unavailable ^a	7.5 (5.1-9.2)
Dose rate at prescription depth (mGy/min)	13 (9.2-33)	16.8 (6.8-48.8)	12.1 (9.2-18.6)

I-125, iodine-125; Ru-106, ruthenium-106.

^a Because of the swift dose fall-off of Ru-106, the dose to the opposite retina cannot be reliably calculated. It is considered exceedingly low.

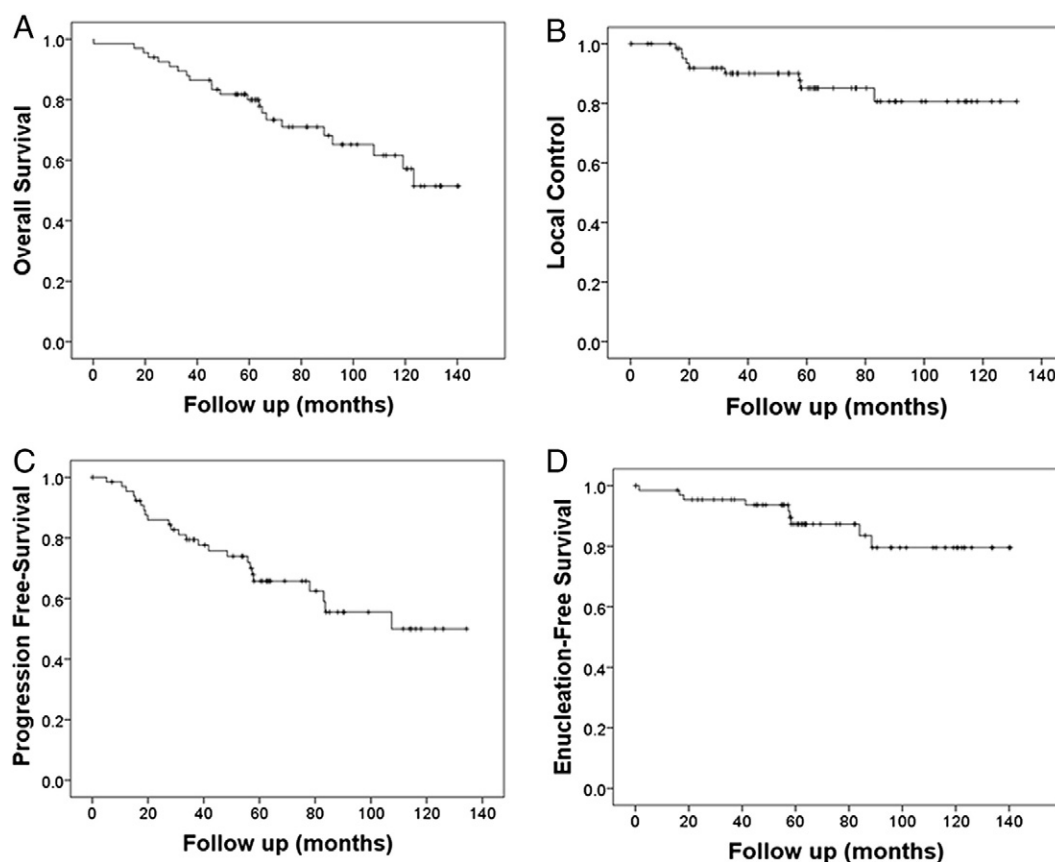


Figure 2 Patients treated with iodine-125 eye plaque brachytherapy. (A) Actuarial overall survival rate, (B) actuarial local control, (C) actuarial progression-free survival rate, and (D) actuarial enucleation-free survival rate.

separation, 6 (9%) were diagnosed with optic neuropathy, 5 (8%) with macular edema, and 5 (8%) were diagnosed with radiation-induced vascular occlusion (defined as either retinal artery or vein occlusion). This last finding is noteworthy because this toxicity was experienced by none

of the patients treated with ^{106}Ru at University of Texas MD Anderson Cancer Center.¹⁹ Forty-three (65%) of the patients experienced a significant adverse effect on their visual acuity. Only 11 of 40 patients with a preimplant visual acuity of ≥ 0.5 ($\geq 20/40$) retained this visual acuity on postprocedure assessment. There was no correlation between radiation retinopathy and inner scleral dose ($P = .33$ by Pearson correlation).

In comparing ^{125}I -treated patients with lesions with apical tumor height ≤ 5 mm ($n = 35$) with those treated with ^{106}Ru ($n = 40$), there was a statistically significant increase in retinopathy ($P = .03$) and incidence of radiation-related cataracts ($P < .01$) in the ^{125}I -treated subset. No other toxicity assessed was significantly different between the 2 subsets (Table 3), although there were trends toward improved visual acuity preservation ($P = .05$), decreased neovascular glaucoma ($P = .06$), and reduced radiation-induced vascular occlusion ($P = .06$) with the use of ^{106}Ru .

Discussion

Ocular melanoma is a relatively rare malignancy with a wealth of prospective data regarding optimal management.

Table 2 Actuarial disease control and ocular preservation outcomes

	2 y (%)	5 y (%)
I-125 patients (N = 67)		
Overall survival	93	80
Local control	91	83
Progression-free survival	85	65
Enucleation-free survival	95	87
Ru-106 patients (N = 40)		
Overall survival	97	92
Local control	97	97
Progression-free survival	97	94
Enucleation-free survival	100	100
I-125 patients with tumor height ≤ 5 mm (N = 36)		
Overall survival	97	89
Local control	94	87
Progression-free survival	91	81
Enucleation-free survival	94	91

I-125, iodine-125; Ru-106, ruthenium-106.

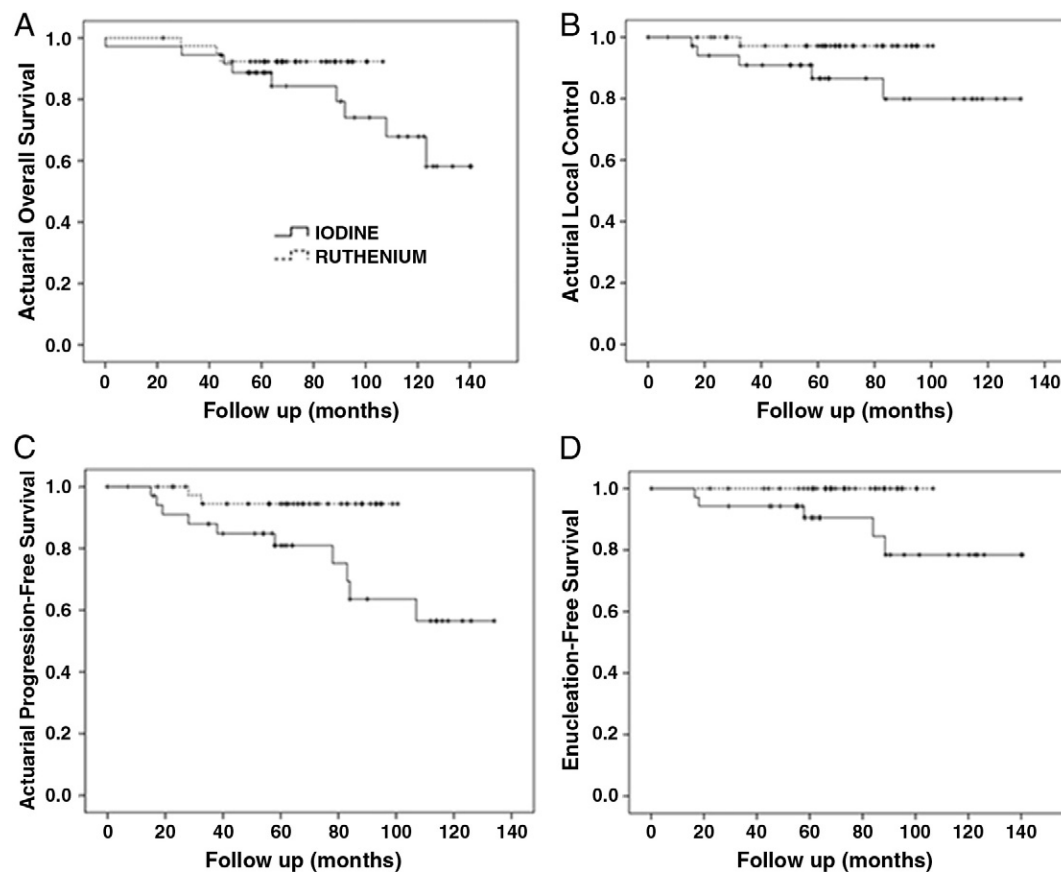


Figure 3 Patients treated with melanomas ≤ 5 mm in apical height treated with iodine-125 and ruthenium-106 (^{106}Ru). (A) Actuarial overall survival rate, (B) actuarial local control, (C) actuarial progression-free survival rate, and (D) actuarial enucleation-free survival rate. Both progression-free survival and enucleation-free survival rates are improved in those patients treated with ^{106}Ru ($P = .02$; $P = .02$).

However, techniques of radiation therapy, namely the optimal radionuclide, have never been directly compared. Our first study objective was to review our single-institutional, single-ophthalmologist experience with ^{125}I brachytherapy in the treatment of ocular melanoma with respect to disease outcomes and radiation-related complications. The median ophthalmologic follow-up duration for patients treated with ^{125}I in our study was 61 months,

facilitating comparison with the 5-year published results from the COMS analyses.¹² The 5-year OS, in our series of 80%, is similar to the 80.7% in the COMS data; the COMS cumulative 5-year enucleation rate was 12%, our population's cumulative enucleation rate was 13.4% with an actuarial EFS of 87%.³ Our cataract rate at 41% is slightly less than the COMS reported rate of 68% at 5 years.²¹ In short, our single-institutional data parallel the

Table 3 Comparison of radiation-associated toxicities in patients treated with iodine-125 and ruthenium-106 in tumors ≤ 5 mm in apical height

Toxicity	No. of affected patients treated with iodine-125 with tumor height ≤ 5 mm (%) (n = 35)	No. of affected patients treated with ruthenium-106 (%) (n = 40)	P value
Radiation retinopathy	26 (74)	20 (50)	.03 ^a
Decreased visual acuity	21 (60)	15 (38)	.05
Cataracts	13 (37)	1 (3)	<.01 ^b
Neovascular glaucoma	3 (9)	0 (0)	.06
Posterior vitreous separation	3 (9)	4 (10)	.83
Optic neuropathy	4 (11)	1 (2.5)	.12
Macular edema	3 (9)	1 (2.5)	.24
Radiation-induced vascular occlusion	3 (9)	0 (0)	.06

^a $P < .05$.

^b $P < .01$.

published findings of the COMS trials with respect to the use of ^{125}I episcleral brachytherapy for the treatment of uveal melanomas.

Our second objective was to use our ^{125}I -treated patients as a control to test the hypothesis that ^{106}Ru , known to have a more limited dose range, may result in decreased toxicity in patients with tumors ≤ 5 mm in height. Using our subset of ^{125}I patients with similarly sized tumors, we observed no difference in OS between those patients treated with ^{125}I and ^{106}Ru ; however we did observe improved PFS in patients who were treated with ^{106}Ru . We additionally identified statistically significant differences in postradiation complications, favoring treatment with ^{106}Ru . Most notably, there were no enucleations in patients treated with ^{106}Ru , and there were significantly lower rates of cataract formation and retinopathy. Although this is not prospective, it is internally well-controlled retrospective data that support the hypothesis that ^{106}Ru eye plaque brachytherapy does result in reduced toxicity with equivalent, if not improved, tumor control in appropriately selected patients. In light of its reintroduction into the US market, this is significant and potentially treatment-altering information.

As with any retrospective approach, there are limitations to the analyses and resultant conclusions. Outcomes and toxicity evaluation are based on chart review and the limitations thereof, although these findings were validated by the ophthalmologist who has implanted and followed all of the patients discussed. We also present our data at a mean ophthalmologic follow-up of 5 years, although our confidence in our findings is increased given the degree of correlation with the results of the large, multicenter COMS study.

Another theoretical limitation of this analysis is that there was a potentially important dosing difference between the 2 radionuclides for tumors ≤ 5 mm in apical height. Per COMS guidelines, the large majority of patients (32 of 36) treated with ^{125}I having tumors with an apical tumor height ≤ 5 mm were prescribed 85 Gy to a depth of 5 mm. In contrast, patients treated with ^{106}Ru (all of whom had tumors ≤ 5 mm in apical height) were treated to a dose of 90 Gy to the apical height itself; hence, the patients treated with ^{106}Ru had lower apical mean doses than those patients treated with ^{125}I . This did alter our practice in the latter portion of this study; for 4 patients treated with ^{125}I for tumors less than 5 mm in height, we did not treat 85 Gy to 5 mm but instead began giving a 1-mm margin to the apical height.

In conclusion, this study not only presents our single institutional experience with ^{125}I episcleral brachytherapy in the treatment of uveal melanoma, but it also presents the first comparative data in support of the use of ^{106}Ru over ^{125}I in the treatment of uveal melanomas ≤ 5 mm in apical height. Although patients treated with both radionuclides experienced excellent rates of tumor control, corroborating previously published findings by other groups, the

improved PFS rate and decreased rate of enucleation and radiation-related toxicity in our single-institutional comparison certainly supports the continued use of ^{106}Ru in appropriately selected patients. For the first time in many years, physicians have a choice of radionuclide for patients with lesions with apical tumor height ≤ 5 mm; these data suggest that both options provide excellent tumor control outcomes and that ^{106}Ru may result in less long-term toxicity. Further study is warranted as more patients are treated in the future.

References

1. Singh AD, Turell ME, Topham AK. Uveal melanoma: Trends in incidence, treatment, and survival. *Ophthalmology*. 2011;118:1881-1885.
2. McLaughlin CC, Wu XC, Jemal A, et al. Incidence of noncutaneous melanomas in the U.S. *Cancer*. 2005;103:1000-1007.
3. Shields CL, Furuta M, Thangappan A, et al. Metastasis of uveal melanoma millimeter-by-millimeter in 8033 consecutive eyes. *Arch Ophthalmol*. 2009;127:989-998.
4. Virgili G, Gatta G, Ciccolallo L, et al. Incidence of uveal melanoma in Europe. *Ophthalmology*. 2007;114:2309-2315.
5. Singh AD, Bergman L, Seregard S. Uveal melanoma: Epidemiologic aspects. *Ophthalmol Clin North Am*. 2005;18:75-84, viii.
6. Eskelin S, Kivela T. Mode of presentation and time to treatment of uveal melanoma in Finland. *Br J Ophthalmol*. 2002;86:333-338.
7. Bergman L, Nilsson B, Lundell G, et al. Ruthenium brachytherapy for uveal melanoma, 1979-2003: Survival and functional outcomes in the Swedish population. *Ophthalmology*. 2005;112:834-840.
8. Virgili G, Gatta G, Ciccolallo L, et al. Survival in patients with uveal melanoma in Europe. *Arch Ophthalmol*. 2008;126:1413-1418.
9. Garretson BR, Robertson DM, Earle JD. Choroidal melanoma treatment with iodine 125 brachytherapy. *Arch Ophthalmol*. 1987;105:1394-1397.
10. Packer S, Rotman M. Radiotherapy of choroidal melanoma with iodine 125. *Int Ophthalmol Clin*. 1980;20:135-142.
11. Collaborative Ocular Melanoma Study Group. The COMS randomized trial of iodine 125 brachytherapy for choroidal melanoma: V. Twelve-year mortality rates and prognostic factors: COMS report no. 28. *Arch Ophthalmol*. 2006;124:1684-1693.
12. Diener-West M, Earle JD, Fine SL, et al. The COMS randomized trial of iodine 125 brachytherapy for choroidal melanoma, iii: Initial mortality findings. COMS report no. 18. *Arch Ophthalmol*. 2001;119:969-982.
13. Sieving PA. Fifteen years of work: The COMS outcomes for medium-sized choroidal melanoma. *Arch Ophthalmol*. 2001;119:1067-1068.
14. Rivard MJ, Coursey BM, DeWerd LA, et al. Update of AAPM task group no. 43 report: A revised AAPM protocol for brachytherapy dose calculations. *Med Phys*. 2004;31:633-674.
15. Sealy R, Buret E, Cleminshaw H, et al. Progress in the use of iodine therapy for tumours of the eye. *Br J Radiol*. 1980;53:1052-1060.
16. Fontanesi J, Meyer D, Xu S, et al. Treatment of choroidal melanoma with i-125 plaque. *Int J Radiat Oncol Biol Phys*. 1993;26:619-623.
17. Melia BM, Abramson DH, Albert DM, et al. Collaborative ocular melanoma study (COMS) randomized trial of i-125 brachytherapy for medium choroidal melanoma. I. Visual acuity after 3 years COMS report no. 16. *Ophthalmology*. 2001;108:348-366.
18. Wilkinson DA, Kolar M, Fleming PA, et al. Dosimetric comparison of ^{106}Ru and ^{125}I plaques for treatment of shallow (≤ 5 mm) choroidal melanoma lesions. *Br J Radiol*. 2008;81:784-789.

19. Takiar V, Gombos DS, Mourtada F, et al. Disease control and toxicity outcomes using ruthenium eye plaque brachytherapy in the treatment of uveal melanoma. *Pract Radiat Oncol.* 2014;4: e189-e194.
20. Chiu-Tsao ST, Astrahan MA, Finger PT, et al. Dosimetry of (125) i and (103)pd COMS eye plaques for intraocular tumors: Report of task group 129 by theAAPM and ABS. *Med Phys.* 2012;39: 6161-6184.
21. Collaborative Ocular Melanoma Study Group. Incidence of cataract and outcomes after cataract surgery in the first 5 years after iodine 125 brachytherapy in the collaborative ocular melanoma study: COMS report no. 27. *Ophthalmology.* 2007;114:1363-1371.